

Does Pretraining Provenance Govern Label Efficiency?

A Preliminary Ablation for Tibial Plateau Fracture Detection

Iván de Jesús López Soto 

Maestría en Ciencias de la Computación
Tecnológico Nacional de México en León
León, Guanajuato, México
lopezsotoivan01@gmail.com

Carlos Lino-Ramírez 

División de Estudios de Posgrado e Investigación
Tecnológico Nacional de México en León
León, Guanajuato, México
carloslino@leon.tecnm.mx

Víctor Manuel Zamudio-Rodríguez 

División de Estudios de Posgrado e Investigación
Tecnológico Nacional de México en León
León, Guanajuato, México
vic.zamudio@leon.tecnm.mx

David Asael Gutiérrez-Hernández 

División de Estudios de Posgrado e Investigación
Tecnológico Nacional de México en León
León, Guanajuato, México
david.gutierrez@leon.tecnm.mx

Juan Martín Carpio-Valadez 

División de Estudios de Posgrado e Investigación
Tecnológico Nacional de México en León
León, Guanajuato, México
juanmartin.carpio@leon.tecnm.mx

Abstract—Recent work assumes that radiology-specific pretraining (RadImageNet) outperforms generic ImageNet pretraining in small orthopaedic cohorts, but this assumption has not been tested under controlled, patient-grouped evaluation. We ablate pretraining provenance — ImageNet, RadImageNet, and random initialization — for tibial plateau fracture (TPF) detection on the public PlaTiF dataset (419 radiographs from 186 patients; 190 knee units, 67% fracture-positive), holding the backbone (ResNet-50), folds, optimizer, augmentation, and tuning budget identical across all three initializations so that provenance is the only manipulated variable. Evaluation is patient-grouped (StratifiedGroupKFold); predictions are made per knee by averaging view logits; and sources are contrasted by paired Δ AUROC with a cluster bootstrap that resamples whole patients. Across label fractions (10/25/50/100%) and five seeds, RadImageNet and ImageNet are statistically indistinguishable (Δ AUROC = +0.015, 95% CI [-0.11, +0.14]), and provenance does not affect label efficiency at any budget (all per-fraction intervals cross zero). Both pretrained sources exceed random initialization only once labels are sufficient (Δ AUROC $\approx$ +0.15 at 100%, interval excludes zero), with no benefit at $\leq$ 25%, where every initialization remains near chance (AUROC $\approx$ 0.5). Peak AUROC is 0.68. We read these preliminary observations cautiously: in this single-cohort setting we simply did not find the assumed advantage of radiological pretraining, and pretraining of either kind helped only once labels were sufficient. We adopt this, provisionally, as a working assumption for the design of an in-domain masked-autoencoder protocol we intend to study next, and release fold manifests, out-of-fold predictions, and configurations to invite scrutiny.

Index Terms—tibial plateau fracture, transfer learning, pretraining provenance, label efficiency, RadImageNet, patient-grouped evaluation, musculoskeletal radiography

I. INTRODUCTION

Automated analysis of tibial plateau fractures (TPFs) on plain radiographs has matured rapidly. Multi-centre supervised detectors now reach expert-level sensitivity: van der Gaast *et al.* report a detection sensitivity of 92.7% on 1,506 radiographs from 753 patients [1], and Huo *et al.* demonstrate a multi-centre detector with external validation and occult-fracture coverage [2]. Fracture *detection* on radiographs is therefore, for practical purposes, an established capability, and is not the contribution we claim.

A separate question governs how such systems are *built* under the data scarcity typical of a single-anatomy injury: what initialization should a small-cohort model start from? A widespread assumption holds that domain-matched pretraining — in-domain radiological weights such as RadImageNet [4] — confers an advantage over generic natural-image (ImageNet) pretraining in small medical cohorts. Recent TPF work adopts this assumption directly: the label-agnostic phenotyping pipeline of Elnakib *et al.* initializes its self-supervised encoder from RadImageNet weights, motivating the choice by the RadImageNet literature and a preliminary run rather than a controlled comparison on the target task [3]. Yet the broader transfer-learning evidence is more equivocal: Raghu *et al.* show that ImageNet transfer often confers little benefit for medical imaging beyond lightweight weight statistics, with far smaller models matching transferred ones [5]. Whether radiological provenance specifically *buys* anything for TPF

detection — and whether any such benefit concentrates where labels are scarce — is untested.

We frame that open cell precisely: *how does pretraining provenance — natural-image (ImageNet) versus general-radiology (RadImageNet) versus random — govern label efficiency for tibial plateau fracture detection under patient-grouped evaluation and paired comparison?* The prior detectors do not ablate provenance or trace label-efficiency curves; the neighbouring TPF pipeline assumes the advantage rather than measuring it. We provide exactly that controlled test. We undertake it as a *preliminary* study with a pragmatic purpose: to inform the initialization and label-budget choices of an in-domain self-supervised (masked-autoencoder) perception protocol we are developing, rather than to settle the question for the field.

Our contributions are three, and deliberately narrow:

- 1) The first controlled ablation of pretraining provenance (ImageNet/RadImageNet/random) $\times$ label efficiency for TPF detection, with patient-grouped partitioning and paired Δ AUROC.
- 2) A direct, preliminary test of the assumed RadImageNet > ImageNet advantage in this regime, which we did not observe at any label fraction.
- 3) A reproducible protocol (versioned fold manifest, QC exclusion log, out-of-fold predictions, per-run configs and seeds) over a public dataset.

II. RELATED WORK

Supervised TPF detection and classification. Beyond the expert-level detectors above [1], [2], work in this space consistently reports that fracture *detection* is reliable while fine-grained Schatzker *classification* lags, tracking the well-documented inter-observer variability of the labelling schemes themselves [1]. This motivates label-agnostic and representation-focused approaches rather than more supervised classifiers, and frames provenance — not architecture — as the lever we study.

Pretraining provenance and self-supervision. RadImageNet provides open ResNet-50 weights trained on 1.35M radiological images and reports 1–9% AUC gains over ImageNet on several musculoskeletal tasks [4]. The Transfusion analysis tempers such claims, attributing much of ImageNet transfer’s apparent value to weight scaling and low-level feature reuse rather than semantic alignment [5]. Self-supervised pretraining has advanced medical image classification in the large-data regime [6], [8], and musculoskeletal-specific self-supervision has been applied to wrist-fracture detection [12] and upper-limb radiograph anomaly detection [13], [14] using corpora such as MURA [10] and FracAtlas [11]. These works establish provenance as a live design axis but do not isolate its effect on TPF label efficiency.

Direct neighbours. Two recent works share our dataset or anatomy but not our question. Elnakib *et al.* perform *unsupervised* phenotype discovery on PlaTiF (SimCLR encoder, UMAP, *k*-means), evaluated by cluster stability and blinded expert coherence rather than any predictive metric;

provenance is assumed, not ablated [3]. Yue *et al.* use a masked autoencoder for TPF *segmentation* in CT under limited annotation — a different modality and task [7]. Our study is the supervised, patient-grouped provenance ablation that this neighbouring work presupposes.

III. MATERIALS AND METHODS

A. Dataset and label unit

We use the public PlaTiF collection [15]: 186 patients, one MATLAB v5 file per patient, each containing a variable number of radiographic views (1–8; 421 views total) with per-view Schatzker labels and clinical metadata. Per-view labels were decoded as fracture-present for Schatzker classes 1–6 and normal for class 7 ($y = 1[\text{label} \neq 7]$), agreeing with the metadata on all 186 patients. Following the correct experimental unit for this problem, prediction and evaluation are performed at the **knee level**: each knee unit is defined programmatically as `knee_id = (patient_id, “fx” if label  $\leq$  6 else “normal”)`, yielding 190 knee units (128 fracture / 62 normal; knee-level prevalence 0.674). Four patients contribute two knee units each (one fractured, one contralateral normal); a guard assertion enforces that this mixed-laterality set matches exactly before any modelling. The negative class is “no tibial plateau fracture”, not “healthy radiograph”.

B. Quality control

QC criteria were frozen before any metric was inspected: minimum resolution, image–mask concordance, near-duplicate detection, and admitted view type. Near-duplicate detection (64-bit perceptual hash, Hamming $\leq$ 5) was restricted to *cross-patient* matches, since folds are patient-grouped and within-patient repeats carry no leakage risk. QC removed 2 views for image–mask shape mismatch and found no cross-patient duplicates; the analysis cohort was 186 patients / 190 knee units / 419 views (Fig. 1 caption). We note this QC is lighter than the eight-step acquisition-artefact protocol used for unsupervised clustering in [3]; Section V addresses why this does not threaten the provenance contrast.

C. Partitioning

We use 5-fold StratifiedGroupKFold over `patient_id`, stratified by patient-level label, so that all views and both knee units of any patient fall in the same fold (zero leakage, verified by an automated test). Within each training fold, an inner patient-grouped split holds out a validation set for early stopping and threshold selection; the test fold is never used for model selection.

D. Model, weights, and preprocessing

All comparators share a single ResNet-50 backbone [9] with a single logit head, trained with BCEWithLogitsLoss using a positive-class weight set to the train-fold negative/positive view ratio. *Provenance is the only difference*: ImageNet (pretrained), RadImageNet (loaded from the RadImageNet ResNet-50 checkpoint [4]), and random initialization. Every weight load asserts the number of matched

tensors; the RadImageNet checkpoint was remapped from its backbone. $\{0, 1, 4, 5, 6, 7\}$ naming to torchvision conv1/bn1/layer1..4 names and required exactly 318 loaded tensors or aborted (the reinitialized head being the only uninitialized module).

Each view’s image was intensity-normalized by its 1st–99th percentiles to $[0, 1]$, bilinearly resized to 224×224 , replicated to three channels, and standardized with ImageNet statistics — identically across all comparators. No augmentation was used: a controlled augmentation variant (flip, small affine, mild brightness/contrast, head dropout) reduced out-of-fold performance and widened the confidence intervals on this small cohort, so it was not adopted. This is a supervised design choice and is orthogonal to self-supervised pipelines, where augmentation is the learning signal rather than an optional regularizer.

E. Training

Identical for all comparators: Adam ($\text{lr} = 3 \times 10^{-4}$, weight decay 1×10^{-4}) [16], batch size 32, up to 30 epochs with early stopping (patience 7) on inner-validation AUROC, restoring the best checkpoint. The per-fold seed was identical across comparators, matching folds, batch order, and head initialization so that only provenance varies.

F. View-to-knee aggregation and metrics

Out-of-fold (OOF) view logits were aggregated to the knee level by the *mean of logits* within each `knee_id`, a rule frozen before any metric was computed; the two knee units of a mixed patient were never averaged together. The primary metric is OOF knee-level AUROC. Provenance contrasts are quantified as ΔAUROC with a paired cluster bootstrap resampling whole patients (2,000 resamples; each patient carries its 1–2 knee units together), reporting the observed effect, 95% percentile interval, and a two-sided bootstrap p as a descriptive summary. We report effect sizes with intervals rather than binary significance, and also report AUPRC and prevalence. Calibration and Brier score are not claimed here.

G. Multi-seed and label-efficiency protocol

Robustness was assessed over independent seeds: the 100% contingency used 5 seeds (1337, 2024, 7, 42, 2718) and the label-efficiency curve used 3 seeds (1337, 2024, 7). Absolute AUROC is reported as mean $\pm$ SD across seeds; ΔAUROC intervals pool the per-seed bootstrap distributions. For the curve, nested class-stratified subsets of the *training* patients ($10\% \subset 25\% \subset 50\% \subset 100\%$) were drawn per fold and seed and were identical across comparators, while validation and test sets were unchanged — isolating the provenance $\times$ fraction interaction. We version the fold manifest, QC log, per-run config with seed, OOF knee predictions, and software environment.

IV. RESULTS

A. Provenance at full label budget

Table I reports OOF knee-level performance at 100% labels over 5 seeds. RadImageNet and ImageNet are close in both

TABLE I
PROVENANCE AT 100% LABELS (5 SEEDS; OOF KNEE-LEVEL, MEAN $\pm$ SD).

Initialization	AUROC	AUPRC
RadImageNet	0.679 ± 0.038	0.800 ± 0.024
ImageNet	0.664 ± 0.028	0.784 ± 0.021
Random	0.526 ± 0.020	0.698 ± 0.022

TABLE II
PAIRED ΔAUROC AT 100% LABELS (PATIENT CLUSTER BOOTSTRAP; MEAN $\pm$ SD ACROSS SEEDS; 95% POOLED CI; DESCRIPTIVE p).

Contrast	ΔAUROC	95% CI	p
RadImageNet – ImageNet	$+0.015 \pm 0.036$	$[-0.109, +0.136]$	0.80
RadImageNet – Random	$+0.154 \pm 0.057$	$[+0.009, +0.286]$	0.035
ImageNet – Random	$+0.139 \pm 0.039$	$[+0.011, +0.267]$	0.035

AUROC and AUPRC; random initialization stays near chance in AUROC and near the prevalence floor in AUPRC.

The paired contrasts (Table II, Fig. 2) make the pattern precise. The provenance contrast of interest — RadImageNet minus ImageNet — is centred near zero with an interval spanning both signs. Both pretrained sources clear random initialization by a comparable margin, with intervals excluding zero.

B. Label-efficiency curve and interaction

Table III and Fig. 2 trace performance as the training-label budget shrinks. Two facts stand out. First, the RadImageNet–ImageNet gap does not widen as labels become scarce: the per-fraction interaction is -0.043 (10%), -0.003 (25%), -0.013 (50%), $+0.025$ (100%), and every 95% interval crosses zero (p between 0.53 and 0.99). Provenance never separates. Second, the benefit of pretraining over random is *concentrated at the full budget*: at $\leq 25\%$ of training patients (13–31 patients) all three initializations sit near chance, and the pretrained–random advantage reaches an interval excluding zero only at 100%.

V. DISCUSSION

We did not observe the assumed advantage of radiological pretraining. Across five seeds and four label budgets, RadImageNet and ImageNet are indistinguishable at the knee level in our runs (Tables II, III); the point estimate of their difference is $+0.015$ AUROC with an interval an order of magnitude wider than the effect. We read this cautiously. It is not evidence against the neighbouring pipeline, whose contribution is an *unsupervised* phenotype geometry rather than a predictive comparison [3]; at most it invites a second look at an assumption that recent TPF work adopted from the RadImageNet literature [4]. One plausible reading, consistent with Transfusion’s observation that much of transfer’s benefit is low-level rather than semantic [5], is that if the value of pretraining here is largely well-scaled early filters, a radiological source need not beat a natural-image one on a task whose discriminative cue — cortical disruption around the plateau —

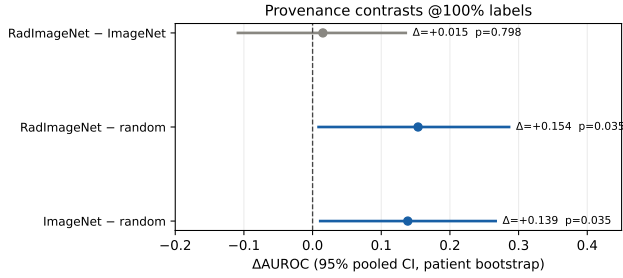


Fig. 1. Forest plot of paired Δ AUROC at 100% labels (patient cluster bootstrap). The RadImageNet–ImageNet interval straddles zero; both pretrained–random intervals exclude it. Cohort after QC: 186 patients / 190 knees / 419 views (2 views excluded for image–mask mismatch).

TABLE III

LABEL-EFFICIENCY CURVE (3 SEEDS; OOF KNEE-LEVEL AUROC, MEAN $\pm$ SD). FRACTIONS ARE NESTED SUBSETS OF *training* PATIENTS.

Fraction (train pts.)	RadImageNet	ImageNet	Random
10% (13)	0.506 $\pm$ 0.022	0.550 $\pm$ 0.045	0.511 $\pm$ 0.024
25% (31)	0.545 $\pm$ 0.030	0.547 $\pm$ 0.029	0.535 $\pm$ 0.038
50% (61)	0.621 $\pm$ 0.027	0.634 $\pm$ 0.011	0.544 $\pm$ 0.035
100% (122)	0.688 $\pm$ 0.006	0.663 $\pm$ 0.031	0.526 $\pm$ 0.007

is a generic edge/texture pattern. We offer this as a conjecture, not a claim.

Where pretraining helps reframes what “label efficiency” can mean. The benefit of any pretraining over random initialization is real but appears only at the full budget ($\Delta \approx 0.15$, interval excluding zero), and vanishes at $\leq 25\%$, where every initialization — pretrained or not — sits near chance (AUROC ≈ 0.5). This is the opposite of the pattern one hopes for from pretraining, which is usually justified precisely by its low-label regime. The most parsimonious reading is that at 13–31 training patients the bottleneck is not the *representation* but the *fine-tuning signal*: there are too few labelled knees for a linear-probe-plus-fine-tune head to locate the decision boundary, regardless of how the backbone was initialized. “Label efficiency” therefore cannot be claimed for any source here; the honest statement is that pretraining does not rescue the scarce-label regime for this task, and provenance does not modulate that.

Why a modest absolute AUROC does not invalidate the ablation. Peak AUROC is 0.68 — modest, and we do not present this as a deployable detector. Crucially, the study’s claim is a *contrast*, and the contrast is internally valid: all three arms share folds, preprocessing, optimizer, augmentation policy, and budget, so any handicap in the pipeline is applied equally and cancels in Δ AUROC. Two choices plausibly cap the absolute level — lighter QC than the eight-step protocol of [3] (we did not exclude wrong-anatomy or post-operative-hardware cases), and full-frame input without an Otsu-based region-of-interest crop centring the plateau. Both would be expected to raise the absolute AUROC of *all* arms together; neither is expected to reverse a provenance ordering whose gap is already +0.015. Establishing that quantitatively — re-

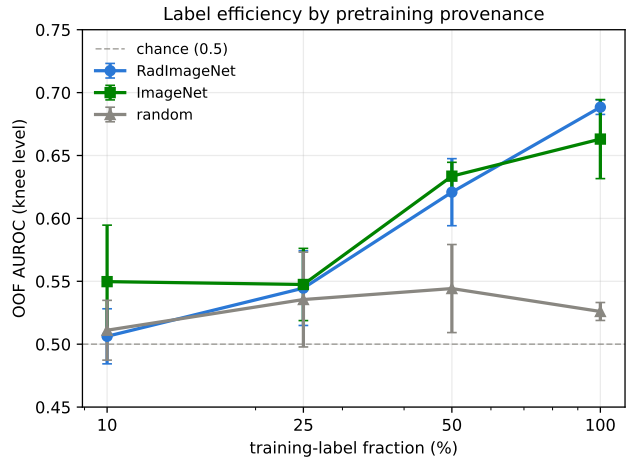


Fig. 2. Label-efficiency curve (OOF knee-level AUROC vs. training-label fraction; 3 seeds). RadImageNet and ImageNet track each other at every budget; both separate from random only as the budget approaches 100%. At $\leq 25\%$ all curves sit near the chance line.

running the ablation on the stricter cohort with plateau-centred crops — is the natural robustness check, and we frame it as future work rather than a threat to the present contrast.

A curious next step, held loosely: in-domain self-supervision. If, in this preliminary setting, neither natural-image nor general-radiology pretraining separates from the other, and if pretraining as a class seems to help only once labels are plentiful, we find ourselves wondering whether the lever worth exploring is a representation matched to *this* anatomy and modality. Self-supervised objectives have advanced medical classification when the pretraining distribution is close to the target [6], [8], and masked-image modelling has shown promising low-annotation transfer for TPF structures in a different modality [7]. This is precisely the direction of our own thesis work, and these experiments were run to inform it. Our curve suggests two simple bars an in-domain masked-autoencoder protocol could be measured against — whether it beats chance at 10–25% labels, or exceeds 0.68 at 100%. We put these forward as questions we are curious to answer, not predictions: it may well be that the limiting factor is the fine-tuning data and task rather than the representation, and we would want to know that too.

Limitations. The cohort is single-institution, 2D radiography, $n = 186$ patients, with no external validation; the absolute AUROC is modest; augmentation and QC were deliberately conservative; and the curve’s low-fraction points rest on 13–31 patients, so their intervals are wide. The findings characterize this task, dataset, and regime, and should not be read as a general verdict on RadImageNet.

VI. CONCLUSION

In this preliminary study, and within its limits, we did not find that pretraining provenance governs label efficiency for tibial plateau fracture detection: in our runs RadImageNet showed no detectable advantage over ImageNet at any label

budget, and both exceeded random initialization only once labels were sufficient. We take this gently, as a working assumption rather than a settled result. For our own immediate purpose — designing an in-domain masked-autoencoder perception protocol — it suggests, provisionally, that an off-the-shelf ImageNet or RadImageNet initialization may be treated as interchangeable, which lets us turn our attention to the question we actually find interesting: whether in-domain self-supervision can move the scarce-label regime that off-the-shelf transfer, here, left near chance. We hold that as a hypothesis to verify, not a conclusion, and release fold manifests, out-of-fold predictions, and configurations so that others — and our future selves — can check and extend it.

REPRODUCIBILITY

Fold manifests, QC exclusion logs, per-run configurations with seeds, out-of-fold knee-level predictions, and recorded software/hardware versions are released with the code. Every weight load is guarded by a tensor-count assertion, and zero patient leakage across folds is enforced by an automated test.

REFERENCES

- [1] J. van der Gaast *et al.*, “Deep learning for tibial plateau fracture detection and classification on plain radiographs,” *The Knee*, vol. 54, pp. 81–89, 2025. doi: 10.1016/j.knee.2025.02.001.
- [2] J. Huo *et al.*, “Deep learning diagnosis of adult tibial plateau fractures: a multicentre study with external validation,” *Radiology Advances*, vol. 2, no. 3, umaf020, 2025. doi: 10.1093/radadv/umaf020.
- [3] M. Elnakib, M. Saad, and A. Al-Kabbany, “Phenotyping tibial plateau fractures via self-supervised learning: a label-agnostic framework with expert validation,” arXiv:2606.17295, 2026. doi: 10.48550/arXiv.2606.17295.
- [4] X. Mei *et al.*, “RadImageNet: an open radiologic deep learning research dataset for effective transfer learning,” *Radiology: Artificial Intelligence*, vol. 4, no. 5, e210315, 2022. doi: 10.1148/ryai.210315.
- [5] M. Raghu, C. Zhang, J. Kleinberg, and S. Bengio, “Transfusion: understanding transfer learning for medical imaging,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2019. doi: 10.48550/arXiv.1902.07208.
- [6] S. Azizi *et al.*, “Big self-supervised models advance medical image classification,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 3478–3488. doi: 10.1109/ICCV48922.2021.00346.
- [7] P. Yue, D. Cai, C. Guo, M. Liu, J. Xia, and Y. Wang, “Learning generalizable features for tibial plateau fracture segmentation using masked autoencoder and limited annotations,” arXiv:2502.02862, 2025. doi: 10.48550/arXiv.2502.02862.
- [8] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, “A simple framework for contrastive learning of visual representations,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2020, pp. 1597–1607. doi: 10.48550/arXiv.2002.05709.
- [9] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016, pp. 770–778. doi: 10.1109/CVPR.2016.90.
- [10] P. Rajpurkar *et al.*, “MURA: large dataset for abnormality detection in musculoskeletal radiographs,” in *Proc. Med. Imaging Deep Learn. (MIDL)*, 2018. doi: 10.48550/arXiv.1712.06957.
- [11] I. Abedeen *et al.*, “FracAtlas: a dataset for fracture classification, localization and segmentation of musculoskeletal radiographs,” *Scientific Data*, vol. 10, 521, 2023. doi: 10.1038/s41597-023-02432-4.
- [12] S. R. Thorat, D. G. Jha, A. K. Sharma, and D. V. Katkar, “Wrist fracture detection using self-supervised learning methodology,” *Journal of Musculoskeletal Surgery and Research*, vol. 8, no. 2, pp. 133–141, 2024. doi: 10.25259/JMSR_260_2023.
- [13] A. Spahr, B. Bozorgtabar, and J.-P. Thiran, “Self-taught semi-supervised anomaly detection on upper-limb X-rays,” in *Proc. IEEE Int. Symp. Biomed. Imaging (ISBI)*, 2021, pp. 1632–1636. doi: 10.48550/arXiv.2102.09895.
- [14] F. Hinterwimmer *et al.*, “From self-supervised learning to transfer learning with musculoskeletal radiographs,” *Current Directions in Biomedical Engineering*, vol. 8, no. 2, pp. 9–12, 2022. doi: 10.1515/cdbme-2022-1003.
- [15] A. Kazemi *et al.*, “PlaTiF: a pioneering dataset for orthopedic insights in AI-powered diagnosis of tibial plateau fractures,” *Scientific Data*, vol. 13, 240, 2026. doi: 10.1038/s41597-026-06560-5.
- [16] D. P. Kingma and J. Ba, “Adam: a method for stochastic optimization,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2015. doi: 10.48550/arXiv.1412.6980.